

HIRDESH SINGH

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PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience in Full Stack Development using JavaScript, TypeScript, React.js, Node.js, Express.js, and REST APIs, building applications with MVC architecture and Git-based version control. Experienced integrating PostgreSQL, MongoDB, and other databases, with a track record of debugging, testing, and shipping features in Agile/Scrum sprints. Built AI/ML-driven automation projects and is eager to apply that experience to AI Agents and workflow automation. A self-starter who takes ownership, collaborates well in teams, and enjoys solving hard problems.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Java, SQL, Python, C++

Frontend: React.js, HTML5, CSS3, Bootstrap

Backend: Node.js, Express.js, REST API Design, MVC Architecture, Spring Boot/MVC (Basic)

Databases: MongoDB, MySQL, PostgreSQL (working knowledge)

Tools: Git, GitHub, Postman, VS Code, Agile/Scrum, CI/CD (familiar), Debugging & Testing

AI & Machine Learning: CNNs, RNNs, LSTMs, YOLOv8, OpenCV — applied to intelligent, agent-style automation projects

Core Computer Science: OOP, Data Structures & Algorithms, DBMS, Operating Systems, SDLC, RESTful Web Services

PROJECTS

Medicare — Online Disease Prediction & Doctor Appointment System

HTML5, CSS3, JavaScript, MongoDB, REST APIs | Live: doctor-meds-frontend.onrender.com

- Built responsive, multi-role (Admin, Doctor, User) full stack UIs for appointment booking and disease-prediction workflows.
- Designed MongoDB database schemas and implemented full CRUD operations for users, doctors, and appointments.
- Built and integrated REST APIs for authentication, scheduling, and role-based access control.
- Applied OOP principles and modular MVC-style design to improve maintainability, scalability, and debugging efficiency.

NextGen Ballot — Online Voting Platform

Node.js, Express.js, MongoDB Atlas, Mongoose, JavaScript | Live: nextgen-ballot.onrender.com

- Developed a secure full stack voting application using Node.js, Express.js, MongoDB Atlas, and Mongoose.
- Implemented OTP authentication, CAPTCHA verification, Aadhaar validation, and session-based login for secure access.
- Designed and tested REST APIs for voter registration, vote casting, result retrieval, and admin operations.
- Deployed the application on Render and managed version control and collaboration using Git and GitHub.

Smart Navigation System for Visually Impaired

Python, YOLOv8, OpenCV, Flask, pyttsx3

- Built a real-time AI-powered automation system using YOLOv8 and OpenCV to detect obstacles from live camera input.
- Designed a modular pipeline (detection → decision-making → voice alerts) integrated with a Flask backend and web interface.
- Collaborated in an Agile workflow, iterating on feedback across sprints to integrate the computer-vision pipeline with the backend and UI.

EDUCATION

Bachelor of Technology in Computer Science and Engineering

VIT Bhopal University | CGPA: 8.47/10 | Expected 2027

ACHIEVEMENTS

- Solved 250+ Data Structures & Algorithms problems across LeetCode and GeeksforGeeks.
- Participated in multiple hackathons, building scalable full stack applications under time constraints.

CERTIFICATIONS

- Generative AI Professional — Oracle
- Oracle Certified Foundations Associate — Oracle
- Google IT Support Professional Certificate — Google
- Bits and Bytes of Computer Networking — Coursera